7. The ip Command Deep Dive (addr, link, route)

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Phase: Phase 3 — Host networking toolkit
Estimated time: 2 days
Slug: `07-ip-command`

Learning objectives

- Use ip addr/link/route fluently
- Bring interfaces up/down (lab)
- Add/delete routes temporarily
- Prefer ip over deprecated ifconfig on Linux

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

Essential `ip` examples (Linux)

```
ip addr show
ip -br addr                          # brief
ip link show
ip link set eth0 up
ip link set eth0 down

ip route show
ip route get 1.1.1.1                 # which path would kernel use?
ip route add 10.50.0.0/16 via 10.0.0.1
ip route del 10.50.0.0/16

ip -s link show eth0                 # counters RX/TX errors
ip monitor all                       # watch changes live
```

macOS cousins

```
ifconfig
route -n get default
networksetup -listallhardwareports
```

Common mistakes

- Making persistent mistakes thinking `ip route add` survives reboot (needs netplan/NetworkManager/systemd-networkd)
- typos adding routes that blackhole traffic
- changing production without console access

Exercises

Work in `exercises/07-ip-command/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/07-ip-command/` until after an honest attempt.

Self-check

1. Command to show routes?
2. What does `ip route get` do?
3. Brief address listing flag?

Further reading

- https://man7.org/linux/man-pages/man8/ip.8.html